

Revised Statement and Proof of the Validity of the Adaptive Conformal Uncertainty Quantifier (ACUQ)

Supplementary note for E3FORMER

Abstract

This note corrects the statement of Theorem 2 of the main paper (Theorem 1 of the previous supplementary material) and gives a complete, self-contained proof. The previous statement was expressed through the event $\prod_{j=1}^H \mathbf{1}\{x_t^j \notin C_t^j\}$, i.e. “*all* H future values fall outside their intervals”. For $H > 1$ this event is *not* the complement of the joint-coverage event of Eq. (1) of the main paper, which requires *all* H future values to fall *inside* their intervals; the complement of the latter is “*at least one* value falls outside”. The error is confined to the way the theorem was *written*: both Algorithm 1 and the proof operate on the joint miscoverage indicator $\text{err}_t = \mathbf{1}\{Y_t \notin C_t\} = 1 - \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\}$. Section 2 states the corrected theorem, Section 3 recalls finite-sample joint validity for a fixed miscoverage rate, and Section 4 proves the corrected long-run guarantee with an explicit $O(1/(\gamma_Q T))$ rate and without any exchangeability assumption. Section 5 discusses the scope of the result and Section 6 lists the resulting edits to the paper.

1 Setting and notation

Let $H \geq 1$ be the forecast horizon and write $[H] = \{1, \dots, H\}$. At each online time step $t = 1, 2, \dots$ the point-forecasting model (the Online Calibrator of E3FORMER) outputs $\hat{Y}_t = (\hat{x}_t^1, \dots, \hat{x}_t^H) \in \mathbb{R}^H$, and the corresponding ground truth is $Y_t = (x_t^1, \dots, x_t^H) \in \mathbb{R}^H$, where x_t^j denotes the realised value at lead time j . The nonconformity score at lead time j is

$$s_t^j := \|x_t^j - \hat{x}_t^j\|, \quad j \in [H]. \quad (1)$$

The quantifier returns a *box* (a hyper-rectangle)

$$C_t = C_t^1 \times C_t^2 \times \dots \times C_t^H, \quad C_t^j = \{y \in \mathbb{R} : \|y - \hat{x}_t^j\| \leq s_t^{*,j}\}, \quad (2)$$

whose half-widths $s_t^{*,1}, \dots, s_t^{*,H}$ are constructed below. Because C_t is a product set,

$$Y_t \in C_t \iff \forall j \in [H] : x_t^j \in C_t^j \iff \forall j \in [H] : s_t^j \leq s_t^{*,j}. \quad (3)$$

The quantity of interest. Eq. (1) of the main paper asks for *joint* (simultaneous, full-horizon) validity, $\mathbb{P}\{\forall j \in [H] : x_t^j \in C_t^j\} \geq 1 - \alpha$. Accordingly we define the *joint miscoverage indicator*

$$\text{err}_t := \mathbf{1}\{Y_t \notin C_t\} = 1 - \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\} = \mathbf{1}\{\exists j \in [H] : x_t^j \notin C_t^j\} \in \{0, 1\}. \quad (4)$$

Eq. (4) is precisely the indicator used in the online update Eq. (18) of the main paper and in line 9 of Algorithm 1, where the *vector* Y_t is tested against the *box* C_t .

Loss set and marginals. Let $L_c = |\mathcal{L}_t|$ and let $\mathcal{L}_t = \{\ell_\tau\}_{\tau=t-L_c+1}^t$ with $\ell_\tau = (\ell_\tau^1, \dots, \ell_\tau^H)$ be the sliding window of past score vectors. $\mathcal{L}_t$ is split uniformly at random into a *marginal* set $\mathcal{L}_{\text{cal}}$ and a *copula* set $\mathcal{L}_{\text{cop}}$. For each $j \in [H]$ the randomised empirical marginal predictive distribution is

$$\hat{F}_j(s) := \frac{1}{|\mathcal{L}_{\text{cal}}| + 1} \left(\tau + \sum_{\ell \in \mathcal{L}_{\text{cal}}} \mathbf{1}\{\ell^j < s\} \right), \quad \tau \sim \text{Unif}(0, 1), \quad (5)$$

and $\hat{F}_j^{-1}(u) := \inf\{s \in \mathbb{R} : \hat{F}_j(s) \geq u\}$ denotes its generalised inverse.

Partial order and empirical copula. For $u, v \in [0, 1]^H$ write $u \preceq v$ iff $u_j \leq v_j$ for all $j \in [H]$. Map every element of the copula set to the unit cube,

$$U := \{u^i\}_{i \in \mathcal{L}_{\text{cop}}}, \quad u^i = (\hat{F}_1(s_i^1), \dots, \hat{F}_H(s_i^H)) \in [0, 1]^H, \quad (6)$$

and define the *empirical copula* (lower-orthant form, with the finite-sample +1 correction)

$$\hat{Q}(u) := \frac{1}{|\mathcal{L}_{\text{cop}}| + 1} \sum_{i \in \mathcal{L}_{\text{cop}}} \prod_{j=1}^H \mathbf{1}\{u_j^i \leq u_j\}, \quad u \in [0, 1]^H. \quad (7)$$

Given a working miscoverage level $a \in \mathbb{R}$, the quantifier selects

$$u^*(a) \in \arg \min_{u \in [0, 1]^H} \sum_{j=1}^H u_j \quad \text{subject to} \quad \hat{Q}(u) \geq 1 - a, \quad (8)$$

and sets $s^{*,j} = \hat{F}_j^{-1}(u_j^*(a))$ for $j \in [H]$, which yields the box (2). Since $\hat{Q}$ is a right-continuous step function that only increases at the observed vectors, the search in (8) may be restricted without loss to the finite grid $\{u^i\}_{i \in \mathcal{L}_{\text{cop}}} \cup \{\mathbf{1}\}$; the minimum is therefore attained, and remaining ties are broken deterministically by lexicographic order.

Remark 1 (Two typographical corrections in the main paper). Eq. (7)–(8) correct Eq. (15)–(16) of the main paper, which were displayed with the reversed comparison $\mathbf{1}\{u_j^i \geq u_j\}$ (an upper-orthant / survival form) together with the constraint $Q(\cdot) \leq \alpha_t$. Since the coverage event (3) is a *lower*-orthant event $u_t \preceq u^*$, the quantity that must be controlled is the lower-orthant empirical copula (7) at level $1 - \alpha_t$, which is exactly the multivariate quantile $Q(U; \alpha)$ used in the proof of the previous supplementary material and in [2]. Appending the point ∞ to $\mathcal{L}_{\text{cop}}$, as done in the paper, contributes 0 to the numerator of (7) and 1 to its denominator; we have therefore absorbed it into the normalisation $|\mathcal{L}_{\text{cop}}| + 1$.

Adaptive miscoverage rate. The working level is itself updated online. With target level $\alpha \in (0, 1)$, step size $\gamma_Q > 0$, initialisation $\alpha_1 = \alpha$, and $C_t = C_t(\alpha_t)$ obtained from (8) with $a = \alpha_t$, the update of Eq. (18) of the main paper reads

$$\alpha_{t+1} = \alpha_t + \gamma_Q(\alpha - \text{err}_t), \quad t = 1, 2, \dots \quad (9)$$

Definition 2 (Saturation conventions). As in adaptive conformal inference [1], the construction (8) is extended to levels outside $(0, 1)$ by

(C1) if $a \leq 0$, or if the constraint in (8) is infeasible on $[0, 1]^H$, set $s^{*,j} = +\infty$ for all j , i.e. $C_t = \mathbb{R}^H$;

(C2) if $a \geq 1$, set $C_t = \emptyset$.

(C1) and (C2) are the multivariate analogues of $\widehat{Q}_t(1-a) = +\infty$ for $a \leq 0$ and $\widehat{Q}_t(1-a) = -\infty$ for $a \geq 1$ in [1, Sec. 2]. They are what makes the argument of Lemma 9 deterministic. Note that the infeasible branch of (C1) is triggered whenever $\alpha_t < 1/(|\mathcal{L}_{\text{cop}}|+1)$, since $\widehat{Q}(\mathbf{1}) = |\mathcal{L}_{\text{cop}}|/(|\mathcal{L}_{\text{cop}}|+1) < 1$, and that this band is genuinely reachable rather than a vacuous edge case: with the settings of the paper, $|\mathcal{L}_t| = 60$ so $|\mathcal{L}_{\text{cop}}| \approx 30$, $\gamma_Q = 0.05$ and $\alpha = 0.1$, the update (9) decreases α_t by $\gamma_Q(1-\alpha) = 0.045$ at every miscovered step, so two consecutive miscoverages starting from $\alpha_1 = 0.1$ give $\alpha_3 = 0.01 < 1/31 \approx 0.032$. Both branches of (C1) return $u^* = \mathbf{1}$ and $C_t = \mathbb{R}^H$, which is the conservative direction and is what Lemma 3 requires.

The two conventions have the following immediate but crucial consequence, which is stated for the *joint* indicator (4).

Lemma 3 (Saturation). *For every t : (i) if $\alpha_t \leq 0$, or if the constraint in (8) is infeasible at level α_t , then $\text{err}_t = 0$; (ii) if $\alpha_t \geq 1$ then $\text{err}_t = 1$.*

Proof. (i) In either case (C1) gives $C_t = \mathbb{R}^H$, hence $x_t^j \in C_t^j$ for every $j \in [H]$, so $\prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\} = 1$ and, by (4), $\text{err}_t = 0$. (ii) By (C2), $C_t = \emptyset$, hence $Y_t \notin C_t$ and $\text{err}_t = 1$. $\square$

2 Corrected statement

Theorem 4 (Long-run full-horizon validity of ACUQ; corrected form of Theorem 2). *Let $\{C_t\}_{t \geq 1}$ be the sequence of prediction boxes produced by the Adaptive Conformal Uncertainty Quantifier (Algorithm 1) with target miscoverage rate $\alpha \in (0, 1)$, initialisation $\alpha_1 = \alpha$, update rate $\gamma_Q > 0$, update rule (9) and conventions (C1)–(C2). Then the empirical frequency of joint miscoverage converges to α :*

$$\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \left(1 - \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\} \right) = \alpha \quad \text{almost surely,} \quad (10)$$

equivalently $\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{\exists j \in [H] : x_t^j \notin C_t^j\} = \alpha$, equivalently $\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\} = 1 - \alpha$,

i.e. the event of Eq. (1) of the main paper occurs with long-run frequency $1 - \alpha$. Moreover the convergence is deterministic with an explicit rate: for every $T \geq 1$,

$$\left| \frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha \right| \leq \frac{\max\{\alpha_1, 1 - \alpha_1\} + \gamma_Q}{\gamma_Q T} = O\left(\frac{1}{\gamma_Q T}\right). \quad (11)$$

No assumption on the data-generating process — in particular neither exchangeability nor stationarity — is required.

Remark 5 (What was wrong, and why it was only the statement). The previous version of the theorem displayed the summand $\prod_{j=1}^H \mathbf{1}\{x_t^j \notin C_t^j\}$. For $H > 1$,

$$\prod_{j=1}^H \mathbf{1}\{x_t^j \notin C_t^j\} \leq \mathbf{1}\{\exists j : x_t^j \notin C_t^j\} = 1 - \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\}, \quad (12)$$

and the inequality is strict as soon as some but not all coordinates are violated; the two events coincide only for $H = 1$. Consequently the previous statement did *not* imply the guarantee of Eq. (1). A sharp counterexample: let $H = 2$ and suppose coordinate 1 is violated at odd t only, coordinate 2

at even t only, and both simultaneously with frequency α . Then the “all-outside” average equals α while the joint miscoverage frequency in (10) tends to 1. The correction is nevertheless confined to the way the theorem was typeset: the object that is actually fed to Eq. (18)/Algorithm 1 and analysed below is the joint indicator (4), obtained by comparing the vector Y_t with the box C_t .

3 Finite-sample joint validity for a fixed level

We first record the guarantee for a *fixed* level $a = \alpha$, which is the copula-conformal ingredient of the construction and follows the classical conformal-prediction framework [3, 2]. It requires exchangeability and is stated here only for completeness; Theorem 4 does *not* rely on it.

Assumption 6 (Exchangeability). The score vectors $\{\ell_i\}_{i \in \mathcal{L}_{\text{cop}}} \cup \{(s_t^1, \dots, s_t^H)\}$ are exchangeable, and their marginal laws are atomless.

Lemma 7 (Validity of the marginal predictive distributions [4]). *For every $j \in [H]$, the randomised distribution $\hat{F}_j$ of (5) is a conformal predictive distribution: for a test score s^j and any $0 < a < 1$, $\mathbb{P}[\hat{F}_j(s^j) \leq 1 - a] = 1 - a$. In particular $\hat{F}_j(s^j) \sim \text{Unif}(0, 1)$.*

Proposition 8 (Joint validity at a fixed level; [2]). *Let $u_t := (\hat{F}_1(s_t^1), \dots, \hat{F}_H(s_t^H))$ and let $u^* = u^*(\alpha)$ be given by (8), with $s^{*,j} = \hat{F}_j^{-1}(u_j^*)$ and C_t as in (2). Under Assumption 6,*

$$\mathbb{P}[Y_t \in C_t] = \mathbb{P}[\forall j \in [H] : x_t^j \in C_t^j] \geq 1 - \alpha. \quad (13)$$

Proof. The proof reduces the *joint box event* to a *dominance event* in the unit cube and then applies the conformal symmetry argument.

Step 1 (reduction). By (3) and $s^{*,j} = \hat{F}_j^{-1}(u_j^*)$,

$$Y_t \in C_t \iff \forall j \in [H] : s_t^j \leq \hat{F}_j^{-1}(u_j^*) \iff \forall j \in [H] : \hat{F}_j(s_t^j) \leq u_j^* \iff u_t \preceq u^*, \quad (14)$$

where the middle equivalence uses that $\hat{F}_j$ is non-decreasing (“ $\Leftarrow$ ” by monotonicity of $\hat{F}_j^{-1}$, “ $\Rightarrow$ ” up to a $\mathbb{P}$ -null set of ties, excluded because the score laws are atomless by Assumption 6). Note that (14) is an equivalence between two *conjunctions over all $j \in [H]$* : it is the full-horizon event of Eq. (1), not a per-coordinate event.

Step 2 (symmetry). By (8), u^* satisfies $\hat{Q}(u^*) \geq 1 - \alpha$, i.e. $\#\{i \in \mathcal{L}_{\text{cop}} : u^i \preceq u^*\} \geq \lceil (1 - \alpha)(|\mathcal{L}_{\text{cop}}| + 1) \rceil$, and u^* is a measurable function of U alone. Writing $Q(U \cup \{\infty\}; \alpha) := u^*$, one has the quantile-stability property

$$u_t \preceq Q(U \cup \{\infty\}; \alpha) \iff u_t \preceq Q(U \cup \{u_t\}; \alpha),$$

because inserting u_t in place of ∞ does not change the selected quantile when u_t is itself dominated by it. The right-hand event states that u_t ranks among the $\lceil (1 - \alpha)(|\mathcal{L}_{\text{cop}}| + 1) \rceil$ “innermost” elements of the augmented set $U \cup \{u_t\}$ with respect to $\preceq$. Under Assumption 6 the law of this rank is invariant under permutations of the augmented set, whence

$$\mathbb{P}[u_t \preceq u^*] \geq \frac{\lceil (1 - \alpha)(|\mathcal{L}_{\text{cop}}| + 1) \rceil}{|\mathcal{L}_{\text{cop}}| + 1} \geq 1 - \alpha. \quad (15)$$

Combining (14) and (15) gives (13). □

4 Long-run validity of the adaptive scheme

Proposition 8 requires exchangeability, which is untenable for non-stationary cloud workloads. This motivates replacing the fixed α by the adaptive level α_t of (9). The resulting guarantee, stated in Theorem 4, is *distribution-free* and requires *no exchangeability*; the only structural requirement is that the construction obey the saturation conventions (C1)–(C2) of Definition 2. We follow the argument of [1], taking care that every step is carried out for the joint indicator (4).

Lemma 9 (Boundedness of the adaptive level). *For every $t \in \mathbb{N}$, $\alpha_t \in [-\gamma_Q, 1 + \gamma_Q]$.*

Proof. Since $\text{err}_t \in \{0, 1\}$ and $\alpha \in (0, 1)$, each increment satisfies

$$|\alpha_{t+1} - \alpha_t| = \gamma_Q |\alpha - \text{err}_t| \leq \gamma_Q \max\{\alpha, 1 - \alpha\} < \gamma_Q. \quad (16)$$

Suppose, for contradiction, that $\alpha_s < -\gamma_Q$ for some $s \in \mathbb{N}$. Since $\alpha_1 = \alpha > 0$ and, by (16), the sequence moves by strictly less than γ_Q per step, there exists $t < s$ with $\alpha_t \in [-\gamma_Q, 0)$ and $\alpha_{t+1} < \alpha_t$. However, $\alpha_t < 0$ implies, by Lemma 3(i), $\text{err}_t = 0$, and therefore

$$\alpha_{t+1} = \alpha_t + \gamma_Q(\alpha - \text{err}_t) = \alpha_t + \gamma_Q\alpha > \alpha_t,$$

a contradiction. Hence $\alpha_t \geq -\gamma_Q$ for all t . Symmetrically, if $\alpha_t > 1$ then Lemma 3(ii) gives $\text{err}_t = 1$ and $\alpha_{t+1} = \alpha_t + \gamma_Q(\alpha - 1) < \alpha_t$, so α_t can never exceed $1 + \gamma_Q$. $\square$

Proof of Theorem 4. Summing the update (9) over $t = 1, \dots, T$ telescopes:

$$\alpha_{T+1} = \alpha_1 + \gamma_Q \sum_{t=1}^T (\alpha - \text{err}_t) = \alpha_1 + \gamma_Q T \alpha - \gamma_Q \sum_{t=1}^T \text{err}_t. \quad (17)$$

Rearranging (17) and dividing by $\gamma_Q T > 0$,

$$\frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha = \frac{\alpha_1 - \alpha_{T+1}}{\gamma_Q T}. \quad (18)$$

By Lemma 9, $\alpha_{T+1} \in [-\gamma_Q, 1 + \gamma_Q]$, and $\alpha_1 = \alpha \in (0, 1)$, so

$$|\alpha_1 - \alpha_{T+1}| \leq \max\{\alpha_1 - (-\gamma_Q), (1 + \gamma_Q) - \alpha_1\} = \max\{\alpha_1, 1 - \alpha_1\} + \gamma_Q. \quad (19)$$

Substituting (19) into (18) yields the rate (11), which holds surely (hence almost surely) for every $T \geq 1$ and tends to 0 as $T \rightarrow \infty$. Therefore $\frac{1}{T} \sum_{t=1}^T \text{err}_t \rightarrow \alpha$ almost surely. Finally, by the definition (4) of err_t ,

$$\frac{1}{T} \sum_{t=1}^T \text{err}_t = \frac{1}{T} \sum_{t=1}^T \left(1 - \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\}\right) = 1 - \frac{1}{T} \sum_{t=1}^T \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\},$$

which gives (10) and the two equivalent formulations, in particular $\frac{1}{T} \sum_{t=1}^T \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\} \rightarrow 1 - \alpha$ almost surely. No property of the joint law of $\{(X_t, Y_t)\}_{t \geq 1}$ was used: the only inputs are $\text{err}_t \in \{0, 1\}$, the update (9), and Lemma 3. $\square$

5 Remarks on scope and variants

Remark 10 (The adaptive argument is indicator-agnostic — and this is why the corrected theorem holds verbatim). The proof of Theorem 4 uses only that (i) the monitored signal is binary and (ii) it saturates consistently with the level, i.e. Lemma 3. Both properties hold for the joint indicator (4): an infinite box covers *all* coordinates, and an empty box covers *none*. Hence the scheme drives to α exactly the frequency of whichever miscoverage event is fed into (9); feeding the joint event $\{Y_t \notin C_t\}$ — as Eq. (18)/Algorithm 1 do — controls the long-run frequency of the full-horizon event of Eq. (1) at $1 - \alpha$. This is also why the correction does not alter the algorithm, the implementation, or any reported number: only the displayed summand of the theorem changes.

Remark 11 (Per-horizon (marginal) variant). If one prefers per-lead-time validity, $\mathbb{P}[x_t^j \in C_t^j] \geq 1 - \alpha$ for each $j \in [H]$ separately, it suffices to run H independent copies of (9) on the marginal indicators $\text{err}_t^j := \mathbf{1}\{x_t^j \notin C_t^j\}$, with levels $\alpha_t^1, \dots, \alpha_t^H$. Lemma 3 and the proof of Theorem 4 apply unchanged to each j , giving $\frac{1}{T} \sum_{t \leq T} \text{err}_t^j \rightarrow \alpha$ a.s. for every j . The joint and marginal targets are genuinely different objectives — by a union bound the joint miscoverage of H marginally valid intervals can be as large as $\min\{H\alpha, 1\}$ — and E3FORMER targets the joint one, which is the relevant notion for provisioning decisions that must hold over an entire scaling window.

Remark 12 (Nature of the guarantee). Theorem 4 is a statement about the *long-run time-average* frequency of joint coverage, with the finite- T deviation bound (11); like all adaptive-conformal guarantees [1] it does not assert per-step marginal or conditional validity, and it is complementary to the finite-sample bound of Proposition 8, which is exact for a fixed level but requires exchangeability. The trade-off governed by γ_Q is visible in (11): a larger γ_Q shrinks the $O(1/(\gamma_Q T))$ deviation but increases the oscillation of α_t , which empirically narrows the average interval width and can pull the finite-horizon coverage away from $1 - \alpha$ — consistent with the sensitivity study of Fig. 8 in the main paper.

6 Resulting edits to the paper

- (i) **Eq. (1) and Definition 3.** Move the indicator outside the probability and disambiguate the lead-time index, writing the joint event as $\mathbb{P}(x_{t+j} \in c_{t+j}, \forall j \in [H]) \geq 1 - \alpha$, i.e. $\mathbb{P}(Y_t \in C_t) \geq 1 - \alpha$, with $C_t = c_{t+1} \times \dots \times c_{t+H}$ a box of H (not $H + 1$) intervals, aligned with the notation of Definition 2 of the main paper.
- (ii) **Theorem 2 (main paper) / Theorem 1 (supplement).** Replace the summand $\prod_{j=1}^H \mathbf{1}\{x_t^j \notin C_t^j\}$ by the joint miscoverage indicator, giving (10); add the finite- T rate (11).
- (iii) **Eq. (18) and Algorithm 1, line 9.** Define $\text{err}_t := \mathbf{1}\{Y_t \notin C_t\} = 1 - \prod_{j=1}^H \mathbf{1}\{x_t^j \in C_t^j\}$ explicitly, so that the monitored event is unambiguously the complement of Eq. (1).
- (iv) **Eq. (15)–(16).** Adopt the lower-orthant empirical copula (7) with the constraint $\hat{Q}(u^*) \geq 1 - \alpha_t$ in (8), matching the multivariate quantile $Q(U; \alpha)$ used in the proof (Remark 1).
- (v) **Supplement.** Restate the intermediate claim “ $\mathbb{P}[y_j \in C_j] \geq 1 - \alpha, \forall j$ ” as the joint statement $\mathbb{P}[Y_t \in C_t] \geq 1 - \alpha$ of Proposition 8, which is what the displayed chain of equivalences (14) actually establishes; add the saturation conventions of Definition 2 and Lemma 3.
- (vi) **Metrics.** State explicitly that the reported average coverage ratio (ACR) is computed with the same indicator as (4), so that the empirical ACR and the theoretical target $1 - \alpha$ refer to the same event.

References

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